Encapsulating SQL Code

Want code blocks, as in other languages

Standard benefits:

- Encapsulation
- Abstraction
- Code reuse
- Maintenance
- Security
- …

Multiple sublanguages

DQL	Data Query Language	Query data
DML	Data Manipulation Language	Modify data
DDL	Data Definition Language	Modify tables, …
TCL	Transaction Control Language	Create transactions
DCL	Data Control Language	Control user access
Procedural language		Functions

Procedural extensions of SQL

- PL/pgSQL – PostgreSQL
- PL/SQL – Oracle, IBM, …
- Transact-SQL – Microsoft, Sybase

Adds variables, assignment, conditionals, loops, begin/end, functions, exceptions, namespaces, security, transactions, …

Yet another full language. We'll focus on core concepts.

Kinds of code blocks – most SQLs

(User def'd) functions

Side-effect-free

(Stored) procedures

Primarily used for side-effects, e.g., updating DB

Triggers

Event handlers

Kinds of code blocks – PostgreSQL

(User def'd) functions

With or without side-effects

Triggers

Event handlers

Except: PostgreSQL 11 adds procedures.
Procedures allow transactions, functions don't.